

$$= \begin{pmatrix} \lambda / \mu_{a_{11}} & \lambda / \mu_{a_{12}} & \dots & \lambda / \mu_{a_{1n}} \\ \lambda / \mu_{a_{21}} & \lambda / \mu_{a_{22}} & \dots & \lambda / \mu_{a_{2n}} \\ \dots & \dots & \dots & \dots \\ \lambda / \mu_{a_{m1}} & \lambda / \mu_{a_{m2}} & \dots & \lambda / \mu_{a_{mn}} \end{pmatrix} =$$

$$= \lambda \begin{pmatrix} \mu_{a_{11}} & \mu_{a_{12}} & \dots & \mu_{a_{1n}} \\ \mu_{a_{21}} & \mu_{a_{22}} & \dots & \mu_{a_{2n}} \\ \dots & \dots & \dots & \dots \\ \mu_{a_{m1}} & \mu_{a_{m2}} & \dots & \mu_{a_{mn}} \end{pmatrix} = \lambda (\mu A) = \mu \lambda \cdot \tau \quad \square$$

8) $\square$

$$\lambda \cdot \tau = 1 \cdot A = 1 \cdot \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} =$$

$$= \begin{pmatrix} 1 \cdot a_{11} & 1 \cdot a_{12} & \dots & 1 \cdot a_{1n} \\ 1 \cdot a_{21} & 1 \cdot a_{22} & \dots & 1 \cdot a_{2n} \\ \dots & \dots & \dots & \dots \\ 1 \cdot a_{m1} & 1 \cdot a_{m2} & \dots & 1 \cdot a_{mn} \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} = A = \mu \lambda \cdot \tau \quad \square$$

9) $\square$

$$\lambda \cdot \tau = 0 \cdot A = 0 \cdot \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix} =$$

$$= \begin{pmatrix} 0 \cdot a_{11} & 0 \cdot a_{12} & \dots & 0 \cdot a_{1n} \\ 0 \cdot a_{21} & 0 \cdot a_{22} & \dots & 0 \cdot a_{2n} \\ \dots & \dots & \dots & \dots \\ 0 \cdot a_{m1} & 0 \cdot a_{m2} & \dots & 0 \cdot a_{mn} \end{pmatrix} = \begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 \end{pmatrix} = \Theta = \mu \lambda \cdot \tau \quad \square$$